

Transfer Learning & Batch Normalization

Neural Networks & Computer Vision Project

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Core Objectives



Task 1: Transfer Learning

Study the effectiveness of pre-trained models (ResNet-18) when working with limited image datasets compared to CNNs trained from scratch.



Task 2: Batch Norm

Analyze how Batch Normalization influences training stability, convergence speed, and final accuracy in deep neural networks (6+ layers).

Task 1

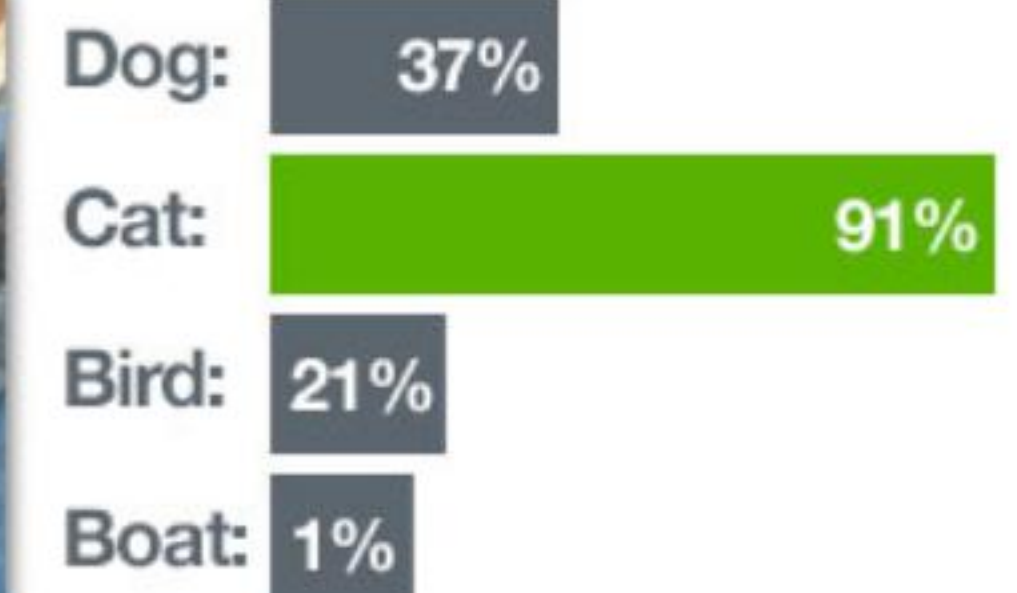
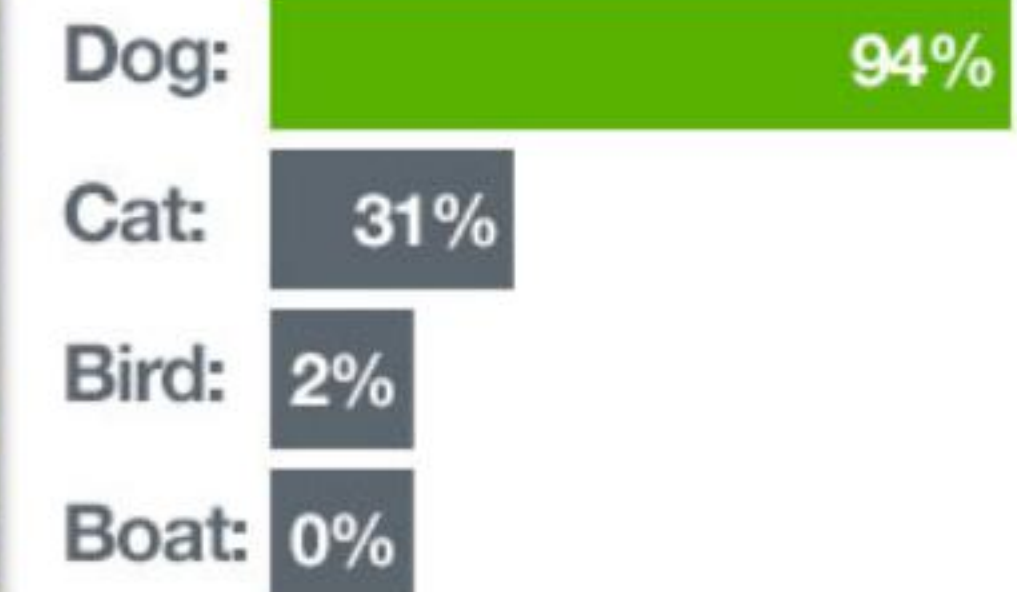
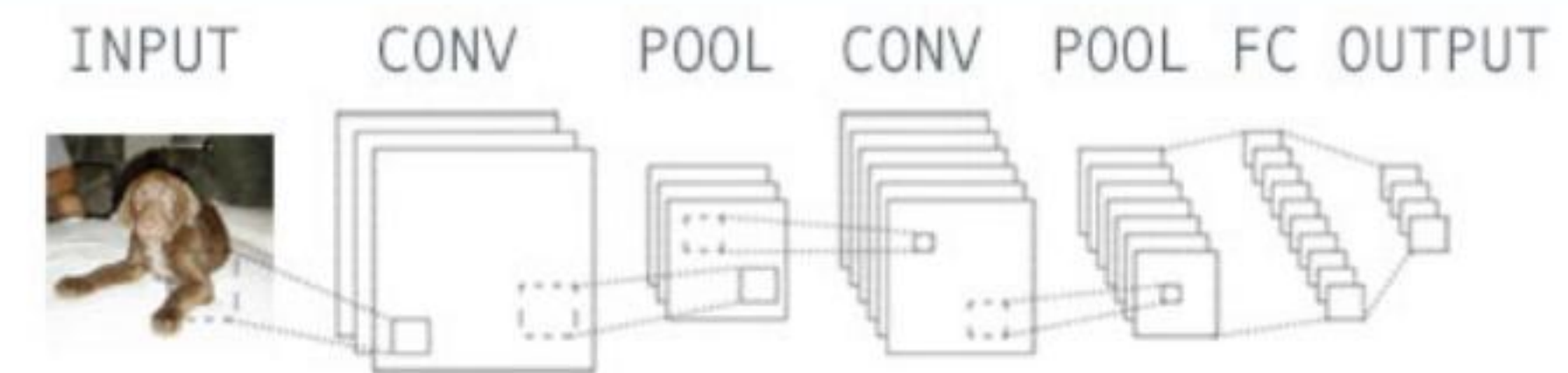
Transfer Learning on Small Datasets

Dataset: Cats vs Dogs Mini

Dataset Statistics

- ✓ **Total Images:** 1000
- ✓ **Split:** 500 Cats / 500 Dogs
- ✓ **Training:** 80% (800 images)
- ✓ **Validation:** 20% (200 images)

Challenge: A dataset this small usually leads to poor feature learning in networks trained from scratch.



Methodology Comparison



CNN Scratch

Simple 3-layer CNN initialized with random weights. Learns features from the 1000 images only.



Frozen ResNet

Pre-trained on ImageNet. Backbone weights are locked; only the final FC layer is trained.



Fine-Tuned

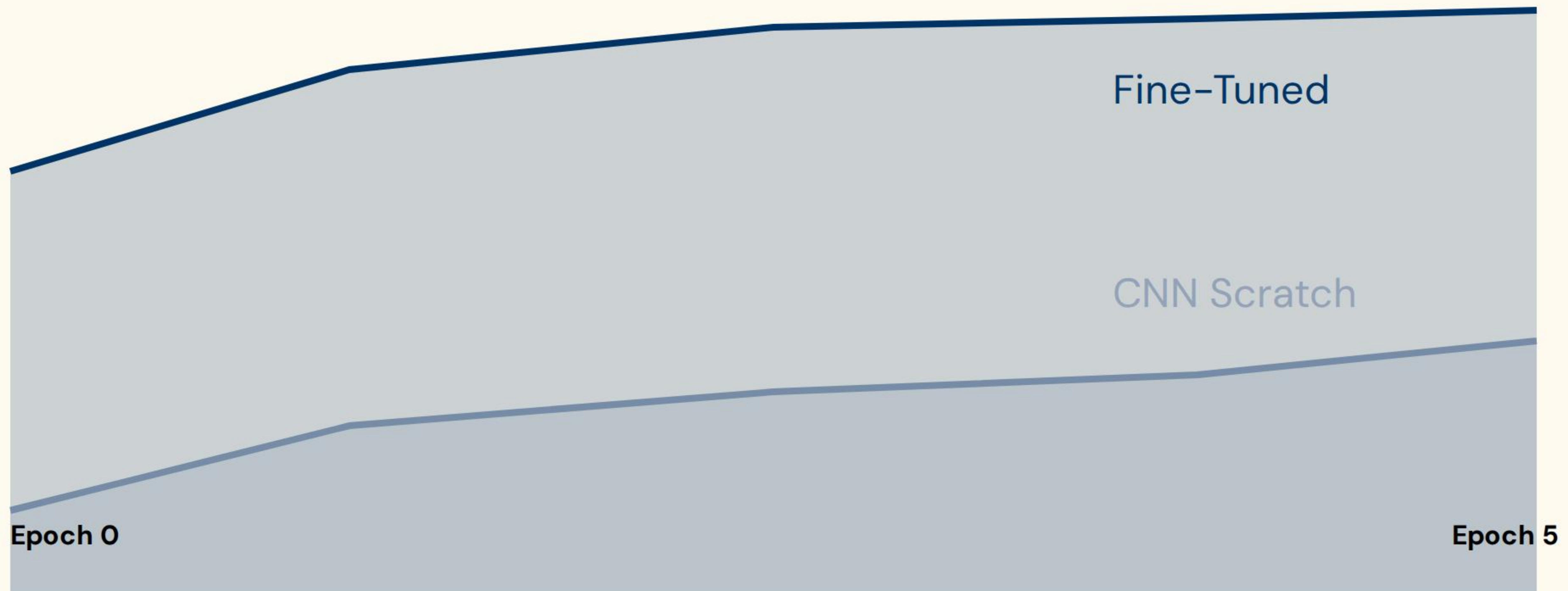
Final convolutional block is unfrozen, allowing the model to adapt specific features to Cats/Dogs.

Accuracy Results (Task 1)



Conclusion: Transfer learning significantly outperforms training from scratch on limited data.

Training Loss Comparison



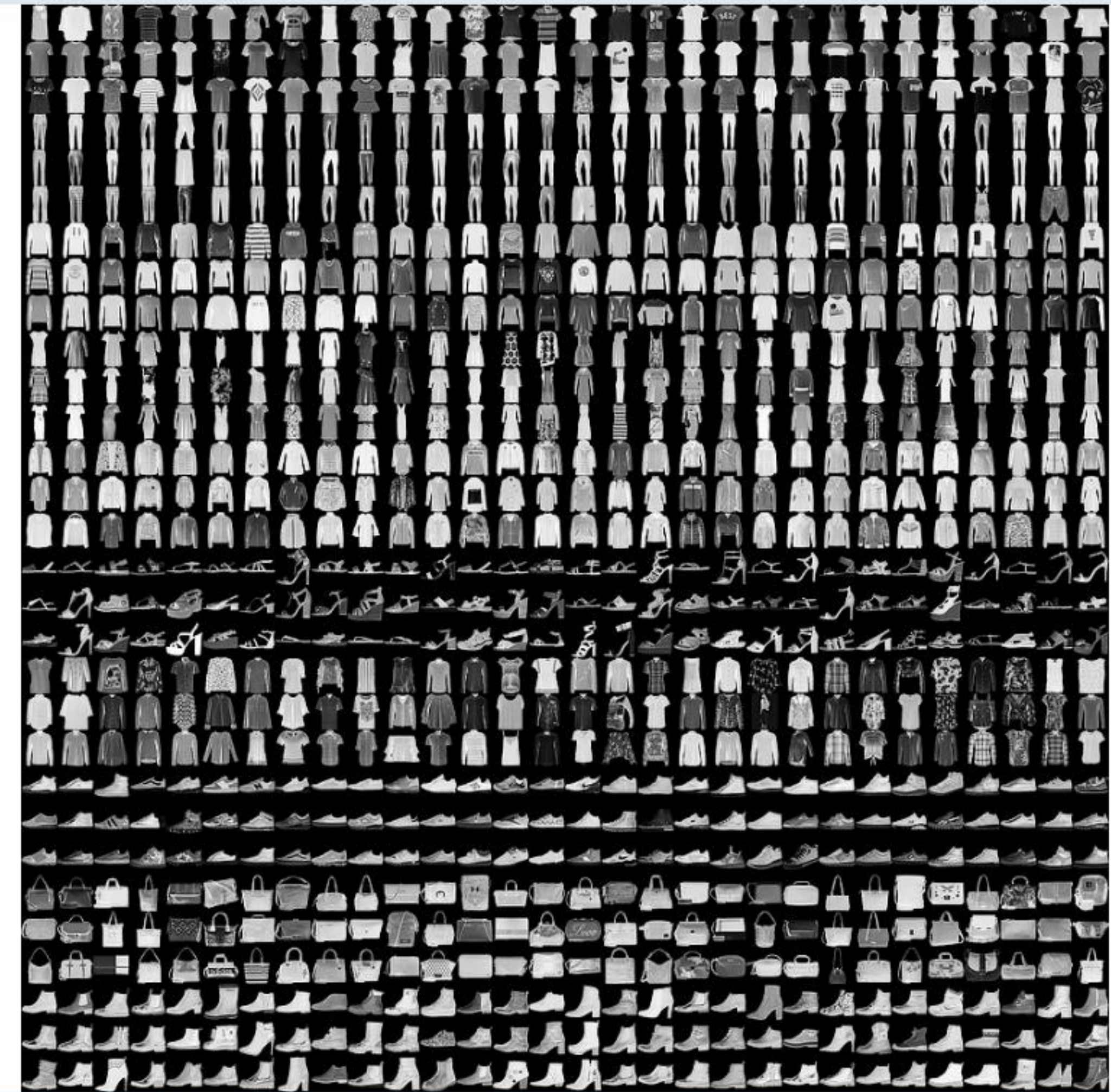
Task 2

Batch Normalization vs No Batch Norm

Dataset: Fashion-MNIST

- ✓ **Content:** 10 clothing categories
- ✓ **Total:** 70,000 Grayscale images
- ✓ **Resolution:** 28x28 pixels
- ✓ **Objective:** Test stability in a deep multilayer perceptron.

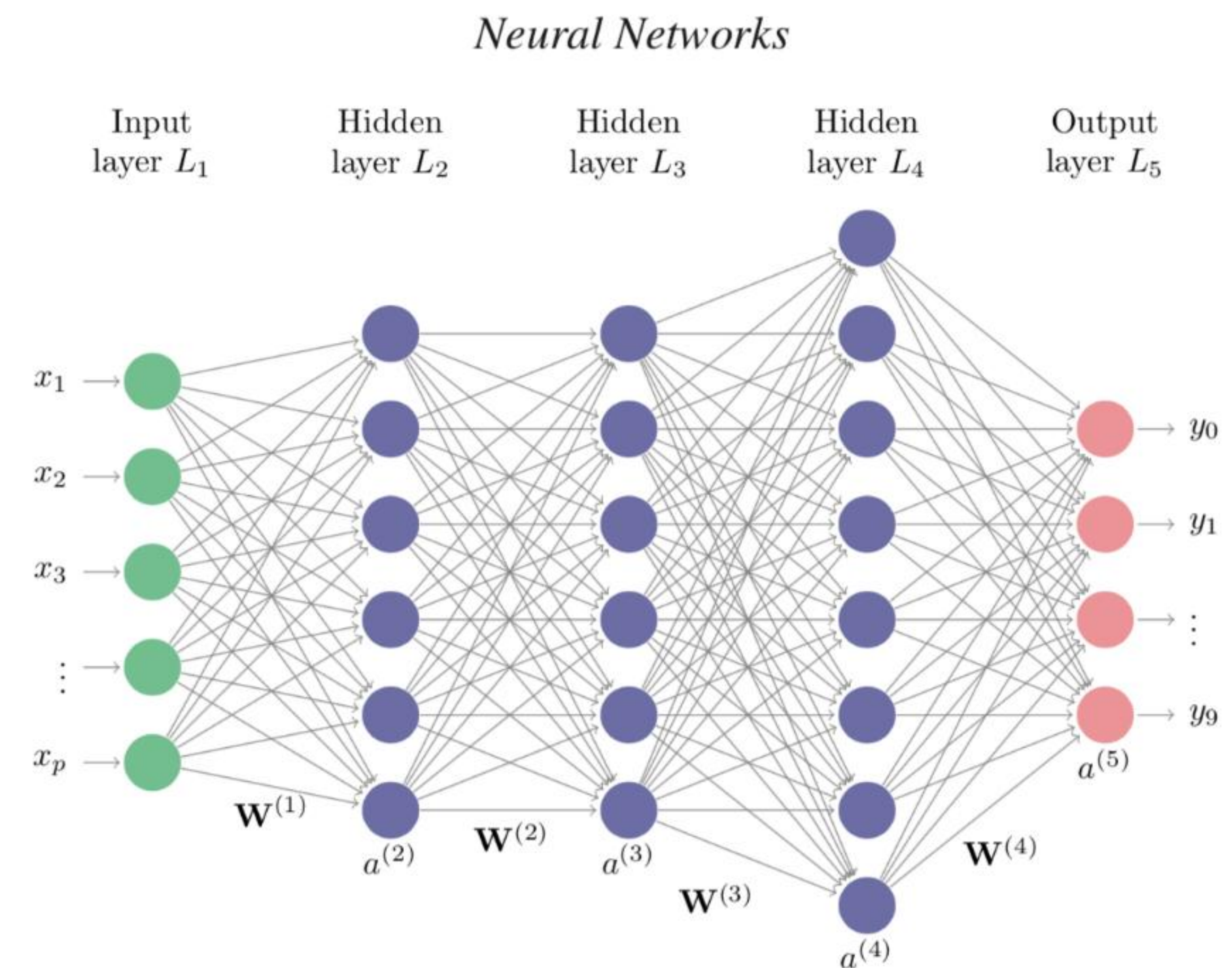
Classes: T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle Boot.



Deep Network Architecture

The "6+ Layers" Setup

- ✓ Input (784) → FC (512) → FC (256)
- ✓ FC (128) → FC (64) → FC (32) → Output (10)
- ✓ **Activation:** ReLU
- ✓ **The Variable:** Inserted *BatchNorm1d* after every hidden layer in Model B.



Batch Norm Accuracy (20 Epochs)

Without Batch Norm

88.75%

With Batch Norm

89.57%

While final accuracy is comparable, Batch Normalization provided much lower **Training Loss** (0.119 vs 0.173) and better **stability** during the initial epochs.

Why Batch Norm Helps?



Faster Convergence

Normalizes activations to prevent vanishing/exploding gradients in deep layers.



Internal Covariate Shift

Reduces the shifting of hidden unit distributions, making optimization easier.



Training Stability

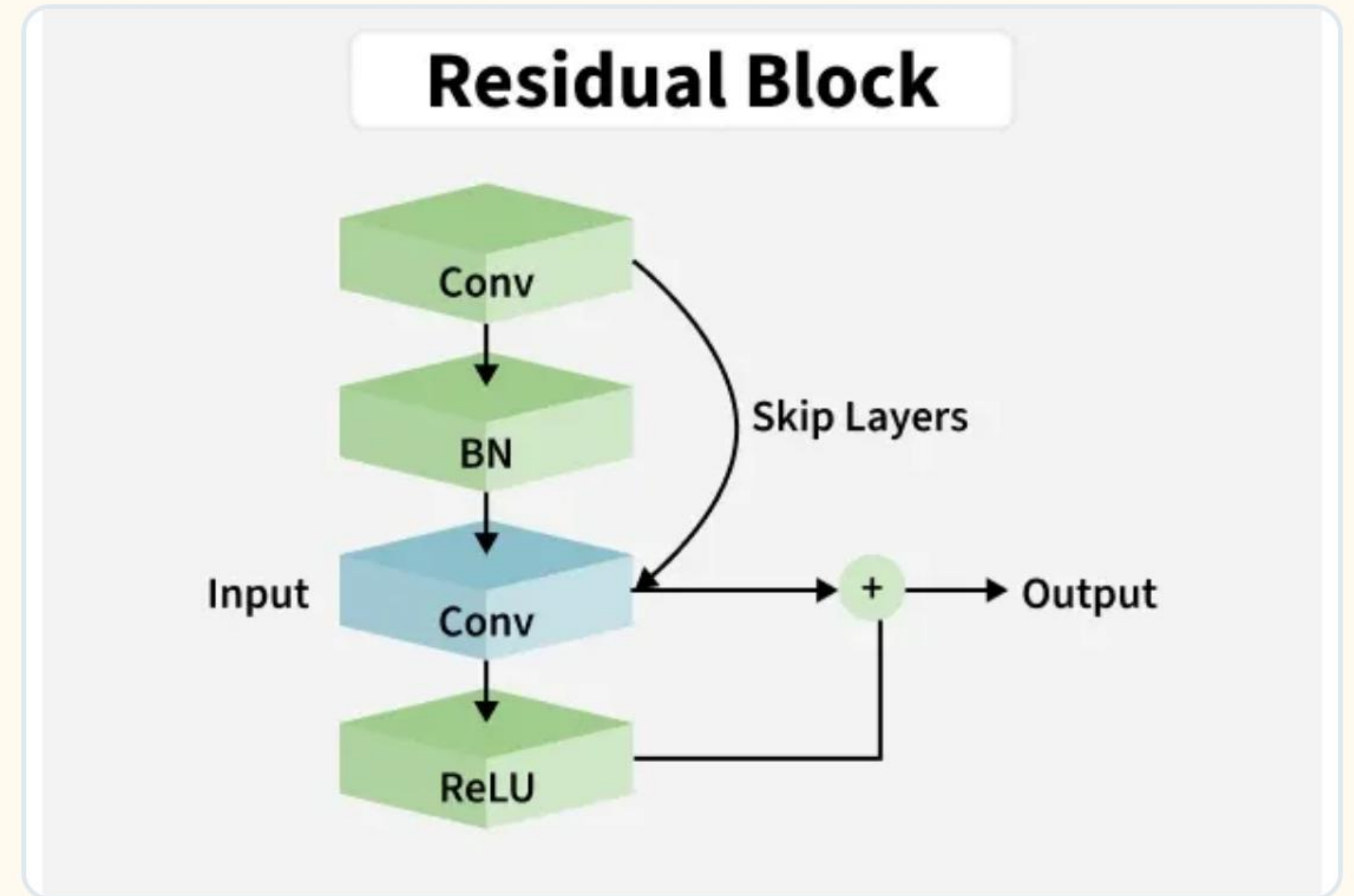
Allows for higher learning rates and makes the network less sensitive to initialization.

Comparative Summary

Experiment	Model Strategy	Final Accuracy	Key Takeaway
Task 1 (TL)	CNN from Scratch	61.5%	Small data ≠ Good feature learning
	Fine-Tuned ResNet	99.5%	Pre-trained weights are superior
Task 2 (BN)	Standard Deep Net	88.75%	Slower convergence & higher loss
	BN Enabled Net	89.57%	Stabilized learning & efficient optimization

Overall Conclusion

- ✓ **Transfer Learning** is essential for real-world projects where data collection is expensive or limited.
- ✓ **Batch Normalization** is a standard tool for making deep networks trainable and robust.
- ✓ The experiments validate that modern architectural improvements (ResNet, BN) solve fundamental optimization hurdles.



Questions?

Thank you for your attention.

Tanush Shoor | SECTION: B | GROUP: 9

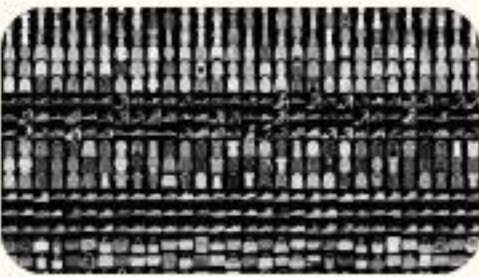
tanush.24bcs10265@sst.scaler.com

Image Sources



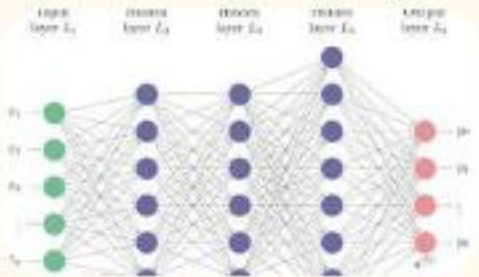
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Source: github.com



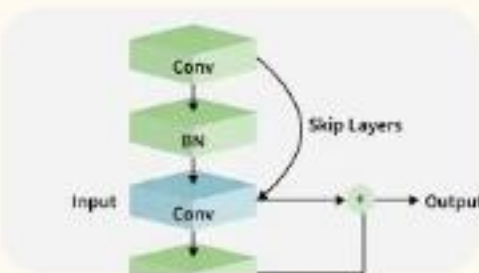
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